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Okonkwo Stanley

Passport: B05150390 | **Work permit:** Nigerian (Nigeria) | **Date of birth:** 03/03/1997 |

Nationality: Nigerian (Nigeria) | **Gender:** Male | **Phone number:** (+234) 08079981193 (Work) |

Email address: okonkwostanley67@Yahoo.com | **Website:** https://github.com/jaguar48/ |

Website: https://github.com/brightstarq | **LinkedIn:**

https://www.linkedin.com/in/stanleyokonkwo/ |

Address: 11 Ugwuaji Road, 400105, Enugu, Nigeria (Home)

ABOUT ME

Environmental geoscientist and GIS researcher with a background spanning hydrogeological field investigation, satellite remote sensing, and ecological modelling. MSc in Ecology and Nature Management (RUDN University, 2026) with thesis research quantifying urban land cover change, green space loss, and flood vulnerability across Lagos State, Nigeria, using multi-temporal Landsat and Sentinel-2 imagery processed in Google Earth Engine. Two peer-reviewed publications. Co-founder of EcoSim, a web-based ecological simulation platform for West African biomes. Professional experience in hydrogeological investigation at the Delta State Ministry of Water Resources and community engagement on land degradation and gully erosion remediation in southeastern Nigeria. Proficient in ArcGIS, QGIS, Google Earth Engine, and Python for geospatial analysis.

WORK EXPERIENCE

CO-FOUNDER AND LEAD DEVELOPER – VERDANT – 13/05/2026 – Current – MOSCOW, RUSSIA

- Co-founded EcoSim and independently built its entire technical architecture, delivering a fully functional web-based ecological simulation platform modelling degradation and restoration dynamics across five distinct West and Central African biomes
- Implemented a quantitative multi-variable ecological health model ($E = 0.28H + 0.30V + 0.22(100\text{-}T) + 0.20B$) governing real-time ecosystem state across five game levels: Desert Bloom (Sahel), Coastal Crisis (Niger Delta), Urban Heat Trap (Lagos), Forest Frontline (Congo Basin), and Planet B (Mars 2140)
- Engineered high-performance 3D terrain rendering using Three.js r167, WebGL GLSL custom shaders, and InstancedMesh for vegetation and geological feature rendering, achieving smooth real-time performance across large scene instances
- Built a full Node.js/Express 5 ESM backend with REST API endpoints, stochastic environmental event engine, and modular architecture designed for future integration with NASA Earth observation and IUCN live biodiversity data feeds
- Developed a complete Canvas 2D prototype prior to the WebGL build, including five playable levels, multiple intervention types, dynamic event systems, community metrics dashboards, and real-time heatmap overlays for soil moisture, surface temperature, and biodiversity

GRADUATE RESEARCHER – INSTITUTE OF ENVIRONMENTAL ENGINEERING, RUDN UNIVERSITY – 09/01/2025 – 07/05/2026 – MOSCOW, RUSSIA

- Independently designed and executed a comprehensive GIS and remote sensing study covering all 20 Local Government Areas of Lagos State across 21 years (2000–2021), integrating six globally validated open-access datasets in a single analytical pipeline
- Processed and analysed multi-temporal imagery from Landsat 7/8, Sentinel-2 MSI, GHSL P2023A, ESA WorldCover 2021, SRTM DEM, and JRC Global Surface Water entirely in Google Earth Engine and Python 3.10, without reliance on pre-processed commercial datasets
- Measured metropolitan drainage density of 0.20 km/km^2 against an internationally recommended standard of 2.0, representing a tenfold deficit, and mapped 3.77 million residents (33.8% of the state population) in high or very high flood vulnerability zones
- Produced six evidence-based policy proposals for sustainable urban environmental management, constituting a directly actionable output for Lagos State planning authorities

- Published two peer-reviewed articles in international journals during the programme period, with DOI-verified outputs in the Journal of Energy Technology and Environment (2025) and Path of Science (2024)
- Presented research findings at two international conferences: the XXXII Lomonosov International Conference at Moscow State University (April 2025) and the II International Youth Scientific Conference on AI Technologies in Science and Education (December 2024)

JUNIOR GEOLOGIST – MINISTRY OF WATER RESOURCES – 07/06/2022 – 14/05/2024 – ASABA, NIGERIA

- Conducted hydrogeological field investigations across erosion-affected and flood-prone communities throughout Delta State, assessing groundwater potential and geological conditions across multiple sites over two years
- Performed systematic borehole logging, aquifer testing, and groundwater level monitoring, generating subsurface datasets used directly in state-level water resource planning and infrastructure investment decisions
- Prepared detailed technical reports on geological conditions, groundwater potential, and water resource management for each investigated site, producing documentation that informed Ministry decisions on borehole siting and water supply development
- Carried out field geological assessments and site evaluations for water resource planning, applying knowledge of sedimentary and basement geology to identify viable aquifer targets in geologically complex areas
- Participated in interdisciplinary water resources management planning committees alongside engineers, environmental officers, hydrologists, and local government representatives, contributing geological expertise to multi-sector decision-making processes

HYDROGEOLOGY TRAINEE – KEY DRILLING & GEOTECHNICAL – 10/04/2020 – 29/10/2021 – AWKA, NIGERIA

- Assisted in drilling operations across multiple sites in Anambra and Ebonyi States, gaining hands-on experience with borehole installation procedures, drilling fluid management, and subsurface formation identification during a 14-month traineeship
- Conducted aquifer testing including pumping tests and recovery tests, collecting time-drawdown data used to calculate hydraulic conductivity and transmissivity parameters for groundwater system characterisation
- Performed systematic geological fieldwork including rock sampling, core logging, and geological mapping across erosion-prone communities in southeastern Nigeria, documenting lithological sequences and structural features
- Supported analysis and interpretation of hydrogeological data using standard geotechnical software, contributing to technical reports on groundwater availability and borehole yield potential
- Participated in hydrogeophysical surveys and borehole site investigations serving rural communities with limited access to safe water supply, gaining direct exposure to the intersection of hydrogeology and community development

EDUCATION AND TRAINING

15/09/2014 – 14/06/2026 Moscow, Nigeria

MASTER'S DEGREE People's Friendship University of Russia

Website https://eng.rudn.ru/ | **Field of study** Ecology and Nature Management | **Level in EQF** EQF level 7

16/12/2015 – 17/12/2019 Awka, Nigeria

BACHELOR OF SCIENCE IN GEOLOGY Nnamdi Azikiwe University Awka

Website nau.edu

12/09/2009 – 15/05/2014 Aba, Nigeria

WEST AFRICAN EXAMINATION COUNCIL Christ the Kind Cathedral Secondary School

PUBLICATIONS

2025

Assessment of Waste Management Issues: A GIS-Based Suitability Analysis

GIS-based spatial suitability analysis of waste management infrastructure, applying remote sensing and geospatial methods to assess environmental site conditions

Oriahki O., Kareem O., Ebite B., Adayi L.E., Okonkwo S.C. (2025)

Authors: Oriahki O., Kareem O., Ebite B., Adayi L.E., Okonkwo S.C. | **Journal Name**: Journal of Energy Technology and Environment | **Volume, Issue and Pages**: Vol. 7, No. 4, pp. 44–52

2024
Geotechnical Investigation of Road Failure Along Ogheghe, South-South, Nigeria

Geotechnical field investigation analysing subsurface conditions contributing to road pavement failure in southern Nigeria

Ojefia F.E., Kalaba F.U., Okafor G.U., Jimoh W.S., Osunbor J.E., Okonkwo S. (2024)

Authors: Ojefia F.E., Kalaba F.U., Okafor G.U., Jimoh W.S., Osunbor J.E., Okonkwo S. | **Journal Name**: Path of Science | **Volume, Issue and Pages**: Vol. 10, No. 9 | **Publisher**: Path of Science

CONFERENCES AND SEMINARS

01/04/2025 – 30/04/2025 Moscow, Russia
XXXII International Lomonosov Conference on Students, Postgraduates and Young Scientists

Oral presentation in the Biology — Environmental Protection subsection titled "Harnessing AI, IoT, IT, GIS, and Community Engagement for Gully Erosion Monitoring and Remediation." Selected via competitive peer review.

01/12/2024 – 31/12/2024 Moscow, Russia
II International Youth Scientific Conference "AI Technologies in Science and Education"

Oral presentation on AI-driven environmental monitoring applications in ecological research and land degradation assessment.

LANGUAGE SKILLS

Mother tongue(s): **ENGLISH**

PROJECTS

16/05/2026 – CURRENT
Verdant — Web-Based Ecological Simulation Platform for West African Biomes

Co-founded and independently built a web-based ecological simulation platform modelling degradation and restoration dynamics across five West and Central African biomes. Implemented a quantitative multi-variable health model, real-time WebGL terrain rendering with custom GLSL shaders, stochastic environmental event engine, and Node.js/Express 5 backend.

Link ecology-simulator.onrender.com

13/01/2025 – 16/05/2026
Ecological Impacts of Urbanisation on Green Spaces and Drainage Patterns in Lagos State, Nigeria

MSc thesis research integrating six satellite datasets across all 20 LGAs of Lagos State over 21 years using GEE. Key findings: 41.1% urban expansion, drainage density of 0.20 km/km^2 against the recommended 2.0, 3.77 million residents in high flood vulnerability zones, Pearson $r = -0.78$ between imperviousness and vegetation loss.